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Optimization of Haacke's Figure of Merit for Indium Tin Oxide Thin Films

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ABSTRACT

The goal of this project is to determine the optimal process conditions for producing indium tin oxide (ITO) thin films with superior electrical and optical properties by controlling the oxygen (O_2) flow rate during radio-frequency magnetron sputtering and the post-deposition annealing temperature. For this purpose, a systematic experimental plan was designed to quantitatively analyze the film properties as a function of two independent variables. We tried three O_2 flow rates (0, 2.5, and 5.0 sccm) and three annealing temperatures (200, 250, and 300 °C) corresponding to nine different process combinations. The properties of these films were thoroughly evaluated using ultraviolet-visible spectrophotometry for optical transmittance, X-ray diffraction for crystal structure, atomic force microscopy for surface roughness, and the Van der Pauw method for sheet resistance. Importantly, the overall film performance is evaluated using Haacke's figure of merit, which provides a quantitative metric for both optical transmittance and electrical conductivity. Through this systematic evaluation, this study seeks to identify the optimal annealing temperature and O_2 flow rate and to establish a comprehensive understanding in optimizing the ITO film as a transparent conducting material.

Keywords: Indium tin oxide, Transparent conductive oxides, Radio-frequency magnetron sputtering, Oxygen flow, Annealing, Haacke's figure of merit

1. Introduction

Indium tin oxide (ITO) is a widely used material in modern optoelectronics, serving as the transparent conductive electrode in a vast array of devices such as flat-panel displays, touch screens, and solar cells. This functionality arises from a unique combination of physical properties, high electrical conductivity comparable to some metals, and high optical transparency in the visible light spectrum [1–4]. Meanwhile, the performance of ITO thin film is critically dependent on the fabrication process. Pre-process parameters during deposition, such as the oxygen (O_2) partial pressure, and post-process treatments, like thermal annealing, can significantly alter the film's microstructure, defect concentration, and surface morphology. This sensitivity poses a significant challenge to manufacturing, where slight process deviations can lead to inconsistent device performance and compromised reliability.

In this context, scientific understanding of the relationship between process variables and material properties is essential. In this study, we systematically investigated the two most important process parameters: vacuum annealing temperature after deposition, which has the greatest effect on light transmittance of ITO, and O_2 flow rate during radio-frequency (RF) sputtering, which has the greatest effect on conductivity. We varied these parameters and analyzed the resulting changes in the film's properties, such as crystallinity, surface roughness, chemical state, conductivity, and transmittance. The ultimate goal is to identify the optimal process window that maximizes the overall performance of the ITO film. This study not only aims to establish an optimized fabrication recipe but also to elucidate the scientific mechanisms behind the observed changes, thereby providing valuable insight and practical guideline for the development of next-generation transparent electronic devices with superior performance and reliability.

2. Theoretical background

2.1. Electrical properties of indium tin oxide

The conductivity of ITO, a composite of In_2O_3 and SnO_2, originates from two primary sources: O_2 vacancies and extrinsic doping with tin. In general, the electrical properties of indium oxide can be either enhanced or suppressed depending on the choice of dopant [5–8]. For maximizing the conductivity, intentional doping of the In_2O_3 with SnO_2 is known as a proper strategy [1–4]. When a tetravalent tin ion substitutes a trivalent indium ion in the lattice, an extra free electron is donated to the system to maintain charge balance, increasing the carrier concentration and conductivity. However, excessive tin doping can be counterproductive, as it may lead to the formation of stable, non-conductive compounds rather than generating free carriers. Therefore, maintaining an optimal dopant concentration is critical,

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with SnO_2 contents in the range of 5 – 10 % generally reported to maximize conductivity [5,9]. In this work, ITO films doped with 10 % SnO_2 were used.

2.2. Optical and structural properties of indium tin oxide

ITO's transparency is a result of its wide bandgap energy. The bandgaps of In_2O_3 (3.6 – 3.7 eV) and SnO_2 (3.6 – 4.0 eV) are larger than the energy of photons in the visible light spectrum (approximately 1.65 – 3.10 eV). Consequently, visible light does not have sufficient energy to excite electrons from the valence band to the conduction band and thus passes through the material without being absorbed [3,4].

The structural state of the film also plays a crucial role. ITO sputtered at room temperature (RT) is deposited in an amorphous state. Post-deposition annealing provides thermal energy for atoms to rearrange into a more stable, ordered polycrystalline structure, typically a cubic bixbyite structure [10,11]. The improved transparency may be attributed to enhanced crystallinity [1]. However, crystallization can also lead to the grain growth, which may increase surface roughness. Excessive roughness can increase light scattering, creating a trade-off between crystallinity and surface smoothness. The intensity of scattered light (I_s) represents the intensity of light scattered by surface roughness. According to the Debye–Waller factor, the scattered light intensity is proportional to the square of the ratio between surface roughness and the wavelength of the incident light, expressed as:

$$I_s \propto (4\pi\sigma/\lambda)^2 \quad (1)$$

where σ is the root-mean-square (RMS) surface roughness, expressed in nanometers (nm). λ is the wavelength of the incident light, expressed in nm. Although I_s is inherently a dimensional physical quantity representing the absolute scattered light intensity (e.g., W/m^2), in this study it is treated as a normalized relative intensity with respect to the incident light, making it dimensionless.

3. Experimental procedure

3.1. Film fabrication

Substrate cleaning

To remove organic and particulate contaminants, substrates were ultrasonically cleaned in sequence with acetone, isopropyl alcohol (IPA), and deionized (DI) water for 5 min each. The principle of ultrasonic cleaning involves creating and collapsing microscopic bubbles that generate high-pressure waves to dislodge contaminants. After drying with a nitrogen gun, the substrates were then dried on a hot plate at 70 °C.

ITO sputtering

A thin film of ITO (40 – 45 nm) was deposited using an RF sputtering system from an In_2O_3:SnO_2 (90:10 %) target. The detailed process recipe is shown in (Table I). RF sputtering was chosen over DC sputtering as it is suitable for depositing dielectric materials like oxides by preventing charge buildup on the target surface. The process was carried out at a working pressure of 3.0 mTorr with an Ar flow of 45 sccm and a power of 150 W. The key process variable, O_2 flow rate, was set to three levels: 0, 2.5, and 5 sccm.

Post-deposition annealing

The as-deposited films were annealed in a rapid thermal processing (RTP) machine for 1 h to promote crystallization and improve film quality. The annealing process was conducted in a vacuum to prevent unwanted oxidation. The chamber pressure was maintained in the range of 5×10^{-3} Torr, and the temperature varied at 200, 250, and 300 °C. The heating rate was set to 8 °C/s, and the cooling rate was controlled at 20 °C/min.

Electrode formation

For electrical measurements, aluminum (Al) electrodes were deposited on the corners of the ITO films via DC sputtering to ensure good ohmic contact for the Van der Pauw measurement.

Table I. Process recipe and experimental design.

Stage	#	Process name	Equipment	Process conditions
Cleaning	0	Wafer cleaning	Sonicator	Acetone 5 min IPA 5 min DI water Hot plate 70 °C 1 min
Deposition	1	ITO deposition	RF magnetron sputtering	Target: ITO Ar flow: 45 sccm O_2 flow: 0 / 2.5 / 5 sccm Base pressure: 3.5×10^{-6} torr Deposition pressure: 3×10^{-3} torr Temperature: RT Sputter power: 150 W Pre sputtering: 10 min Main sputtering: 5 min
Annealing	2	ITO annealing	RTP system	Annealing time: 1 h Annealing temperature: 200 / 250 / 300 °C
Deposition	3	Al electrode deposition	DC magnetron sputtering	Target: Al Ar flow: 50 sccm O_2 flow: 0 sccm Base pressure: 3.5×10^{-6} torr Deposition pressure: 3×10^{-3} torr Temperature: RT DC power: 400 V × 0.3 A Pre sputtering: 1 min Main sputtering: 10 min

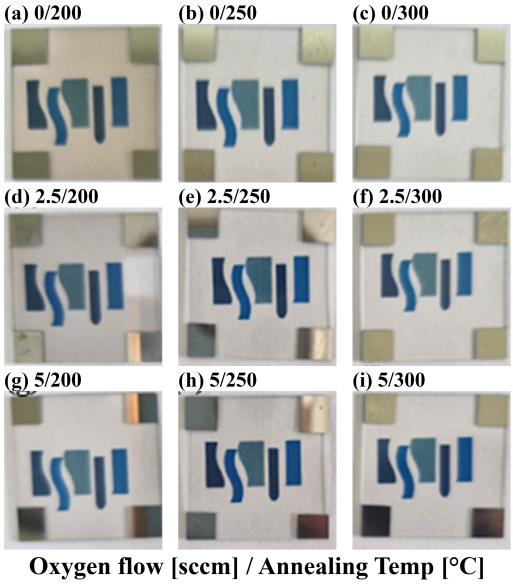


Figure 1. Photograph of fabricated ITO samples arranged by process conditions. (a–c) Images of films deposited at an O_2 flow rate of 0 sccm and annealed at 200, 250, and 300 °C, respectively. (d–f) Images of films deposited at an O_2 flow rate of 2.5 sccm and annealed at 200, 250, and 300 °C, respectively. (g–i) Images of films deposited at an O_2 flow rate of 5 sccm and annealed at 200, 250, and 300 °C, respectively.

Detailed process variables can be found in Table I. This created a 3 × 3 matrix of nine distinct process combinations. For statistical reliability, three samples were made for each combination, totaling 27 samples (Fig. 1).

3.2. Characterization methods

The properties of the fabricated films were comprehensively analyzed.

Optical transmittance

Measured using a ultraviolet-visible spectrophotometer. This technique passes light of a scanned wavelength through the sample and measures the intensity of the transmitted light, governed by the Beer-Lambert Law, to determine transparency across the spectrum.

Crystallinity

Analyzed using X-ray diffraction (XRD), which relies on the constructive interference of X-rays scattered by crystalline planes, as described by Bragg's Law ($n\lambda = 2dsin\theta$). The resulting diffraction pattern provides information on crystal structure, preferred orientation, and grain size via the Scherrer equation.

Surface morphology

We utilized atomic force microscope (AFM) for investigating surface morphology. AFM scans a sharp tip across the sample surface, measuring the forces between the tip and the surface to create a 3D topographic map with nm resolution, from which surface roughness is calculated.

Sheet resistance

Measured using the Van der Pauw method with a 4-point probe system. This is a standard technique to determine the sheet resistance of a uniform, thin sample by applying current between two contacts and measuring the voltage across the other two [16–18].

Haacke's figure of merit – T^{10}/R_s [Ω/sq]

Haacke's figure of merit (FOM) is a widely used quantitative criterion for evaluating transparent conductive thin films. It provides a single parameter that simultaneously reflects both transmittance and electrical conductivity. In general, a higher transmittance and lower sheet resistance indicate better device performance. However, if the FOM is simply calculated as transmittance divided by sheet resistance, a critical issue arises. Transmittance typically varies by about 10 % depending on process parameters, whereas sheet resistance can vary by a factor of two to ten or more. To balance the relative influence of these two properties, Haacke proposed using the 10th power of transmittance, resulting in the expression FOM = T^{10} / R_s [19,20]. T was defined as the single-wavelength transmittance at 550 nm, representing the average transmittance over the visible region (350 – 750 nm).

4. Results and discussion

4.1. Experimental hypothesis

It was hypothesized that as the O_2 flow during sputtering increases, the concentration of O_2 vacancies would decrease, leading to a decrease in conductivity [1,2]. Concurrently, the reduction in vacancies and associated deep-level defect electrons would reduce light absorption, leading to an increase in optical transmittance. It was also hypothesized that as the annealing temperature increases, crystallization would progress, and the concentration of O_2 vacancies may increase (due to the vacuum environment), influencing both conductivity and optical transmittance [1–4]. To quantitatively evaluate both properties simultaneously, Haacke's FOM was adopted for comparison.

4.2. Analysis of optical transmittance

The optical transmittance was evaluated to understand the effects of the process variables. As shown in Figs. 2(a)–(c), the transmittance decreases rapidly from around 310 – 350 nm, which corresponds to the bandgap energy of ITO. Figure 3 is a graph showing the transmission behavior at a 550 nm value, an intermediate value between 350 and 750

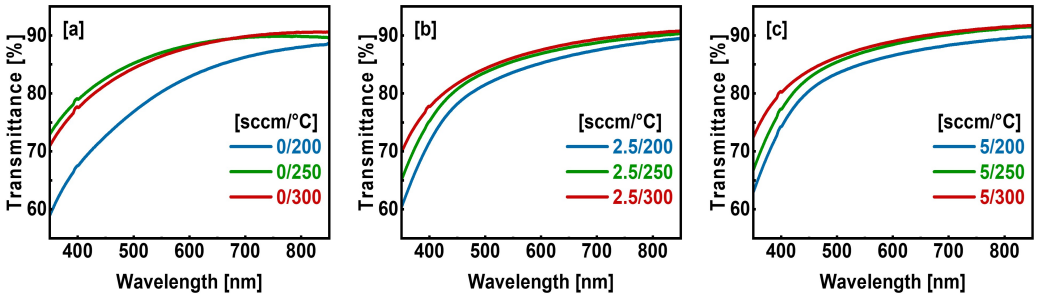


Figure 2. (a–c) Optical transmittance spectra of ITO thin films deposited at O_2 flow rates of 0, 2.5, and 5 sccm, respectively, measured at annealing temperatures of 200 (blue), 250 (green), and 300 (red) °C in the wavelength range of 350 – 850 nm.

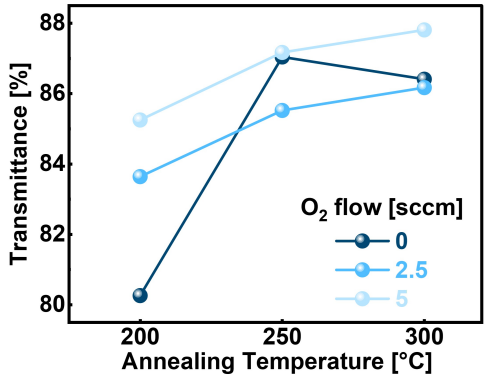


Figure 3. Comparison of the transmittance at 550 nm for all samples as a function of annealing temperature.

nm known as a visible light region. According to Fig. 3, transmittance generally improved with higher annealing temperatures and increased O_2 flow. This aligns with the hypothesis that higher temperatures promote crystallization and higher O_2 flow reduces light-absorbing defects. However, we also observed a key anomaly. For the sample with 0 sccm O_2, the transmittance dramatically increased when annealed at 250 °C but then slightly decreased at 300 °C. It appeared to be the trade-off between crystallization-induced higher transparency, and surface roughness-induced lower transparency. This was covered in Sections 4.5 (XRD analysis) and 4.6 (AFM analysis).

4.3. Van der Pauw results

The sheet resistance was measured using the van der Pauw method to evaluate the electrical conductivity of the ITO thin film. As shown in Fig. 4, the results support the hypothesis that electrical conductivity improves with increasing annealing temperature and decreasing O_2 flow rate.

The improvement in conductivity observed at low O_2 flow rates is attributed to the reduced interaction with O_2 during magnetron sputtering, which leads to an increased concentration of O_2 vacancies and, consequently, enhanced electrical conductivity [21,22]. Regarding the annealing temperature dependence, the rapid decrease in sheet resistance with increasing annealing temperature is attributed to the progression of crystallization in the ITO thin film and the resulting improvement in charge transport. This crystallization behavior is discussed in detail in Section 4.5 through XRD analysis. In addition, since the annealing process was performed under vacuum using an RTP system, an increase in annealing temperature is expected to promote the formation of O_2 vacancies, further reducing the sheet resistance.

A notable trend highlighted in Fig. 4 is an order-of-magnitude decrease in sheet resistance under all O_2 flow conditions as the annealing temperature increased from 200 to 300 °C. In particular, for samples deposited at an O_2 flow rate of 0 sccm, a comparable reduction was achieved with only a 50 °C increase in annealing temperature (from 200 to 250 °C), whereas for samples deposited at higher O_2 flow rates, an increase of 100 °C was required to achieve a similar effect. This distinct behavior is further analyzed in the subsequent XRD results presented in Section 4.5.

4.4. Haacke's figure of merit results

According to the Haacke's FOM shown in Fig. 5, which incorporates both optical transmittance and electrical conductivity, the FOM value increases as the O_2 flow rate during sputtering decreases and the annealing temperature increases. When the annealing temperature increased from 200 to 250 °C, a marked increase in the FOM value was

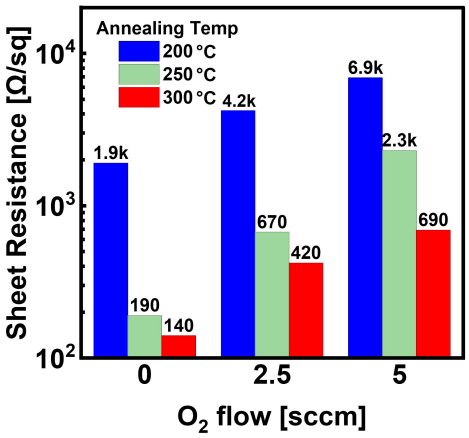


Figure 4. Sheet resistance of ITO thin films depending on O_2 flow rate (0, 2.5, and 5.0 sccm) and annealing temperature (200, 250, and 300 °C).

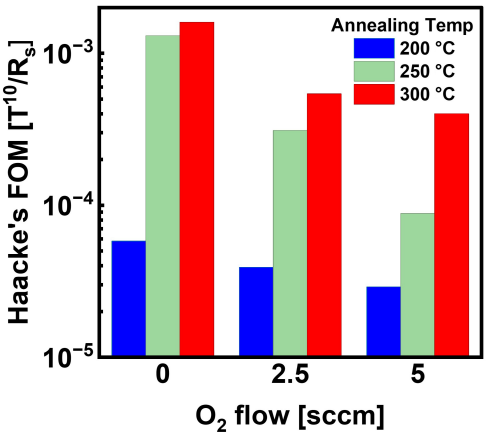


Figure 5. Haacke's FOM of ITO thin films depending on O_2 flow rate (0, 2.5, and 5.0 sccm) and annealing temperature (200, 250, and 300 °C).

observed under all O_2 flow conditions. This trend is consistent with the sheet resistance behavior presented in Fig. 4. At an annealing temperature of 200 °C, the sheet resistance was excessively high for all O_2 flow rates, resulting in a significant degradation of the FOM.

It is noteworthy that the FOM values obtained at annealing temperatures of 250 and 300 °C under an O_2 flow rate of 0 sccm showed no significant difference. As shown in the transmittance behavior in Fig. 3, the optical transmittance at 250 °C is higher than that at 300 °C, while the sheet resistance in Fig. 4 is slightly higher at 300 °C compared to 250 °C, although the difference is not significant. Consequently, these opposing tendencies compensate for each other, leading to comparable FOM values at both annealing temperatures. Overall, the Haacke's FOM analysis confirms that an O_2 flow rate of 0 sccm and an annealing temperature of 250 °C or higher are sufficient to achieve optimized optoelectronic performance of the ITO thin films.

4.5. X-ray diffraction analysis

XRD analysis was performed to investigate the structural changes in the thin films. It is well known that ITO exhibits a characteristic diffraction peak corresponding to the (222) plane [23–25]. In this study, the degree of crystallization was evaluated using the intensity of the (222) peak, and the grain size was calculated and analyzed using the Scherrer equation [25–28]. As shown in Figs. 6(a), 6(d), and

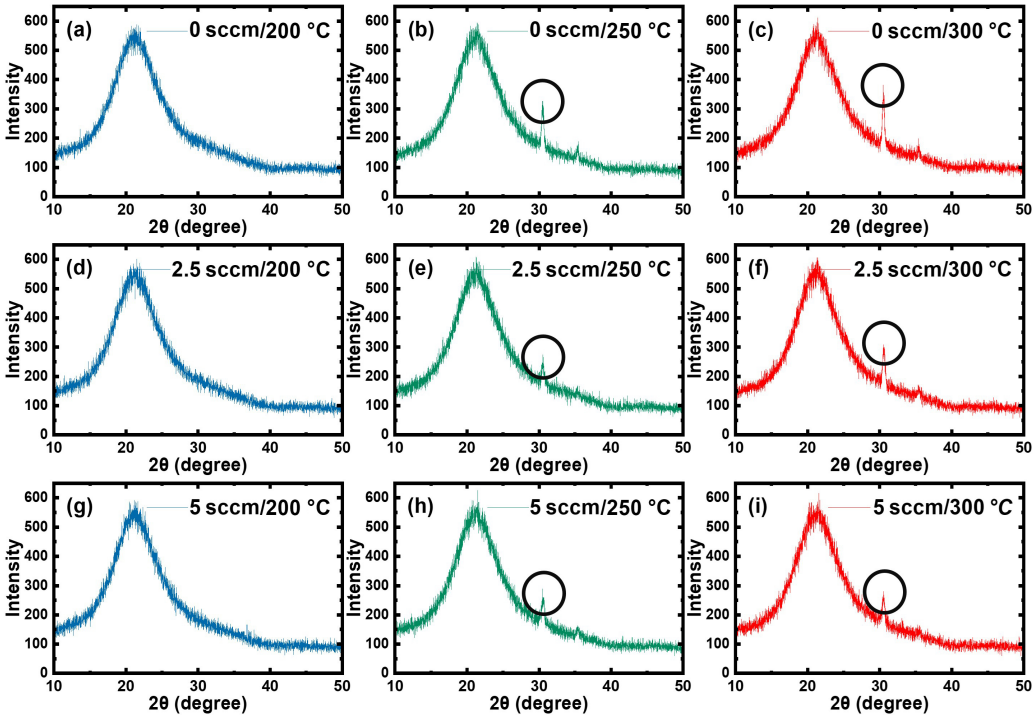


Figure 6. XRD patterns of ITO thin films. (a–c) Patterns of films deposited at an O_2 flow rate of 0 sccm and annealed at 200, 250, and 300 °C, respectively. (d–f) Patterns of films deposited at an O_2 flow rate of 2.5 sccm and annealed at 200, 250, and 300 °C, respectively. (g–i) Patterns of films deposited at an O_2 flow rate of 5 sccm and annealed at 200, 250, and 300 °C, respectively.

6(g), the XRD results indicate that the thin film annealed at 200 °C does not exhibit a distinct crystallization peak, confirming that it remains largely in an amorphous state. Although the crystallization temperature of ITO is generally reported to be approximately 200 °C [3], no noticeable change in the XRD peak was observed at this temperature in the present study. Therefore, this condition serves as a reference point representing the structural state immediately after deposition. Notably, under the annealing temperature of 250 °C, the sample prepared at 0 sccm exhibited a very high (222) peak intensity of 129.6, as summarized in Table II. This value is approximately twice that of the samples prepared at 2.5 sccm (63.4) and 5 sccm (74.2) under the same annealing conditions, indicating significantly enhanced crystallinity. In addition to crystallinity, the grain size was estimated using the Scherrer equation Eq. (2):

$$D = (K\lambda) / (\beta \cos\theta) \quad [\text{nm}] \tag{2}$$

where D is the mean grain size [nm], K is the dimensionless shape factor (typically 0.9), λ is the incident X-ray wavelength [nm], β is the full width at half maximum of the diffraction peak corrected for the instrumental width magnification, and θ is the Bragg angle. The D was calculated using the Scherrer equation, with λ, β, and θ extracted from the XRD data shown in Fig. 6. The calculated values are summarized in Table III.

As listed in Table III, the 0 sccm sample annealed at 250 °C exhibited the largest grain size of 27.2 nm. This result indicates that grain growth was most effective under these conditions, suggesting that both the O_2 flow rate and annealing temperature play critical roles in determining the crystal structure of ITO thin films.

The trend of XRD peak intensity and extracted grain size suggests possible scenario in optical transmittance. The peak intensity increases with annealing temperature, reflecting improved crystallinity, which is often considered to be closely associated with improved optical transmittance [3,4,24,25]. In addition, since grain boundaries may act as major sites for light scattering and absorption, an increase in average grain size can lead to a decrease in the total grain boundary area, thereby enhancing optical transmittance [3,4]. For example, the 0 sccm sample annealed at 250 °C exhibited both the highest (222) peak intensity and the largest grain size, indicating a substantial reduction in grain boundary density within the film. As expected, there was a rapid increase in transmittance when the annealing temperature increased from 200 to 250 °C. In contrast, the sample with O_2 flow rate of 2.5 sccm, annealed at 250 °C exhibited the smallest grain size of 17.6 nm, as shown in Table III. This sample exhibited relatively lower transmittance observed compared to other samples under the same temperature conditions. It should be noted that, however, the slight decrease in transmittance observed for the 0 sccm sample at 300 °C cannot be

Table II. XRD peak height (Max height).

Max height	Annealed at 200 °C	Annealed at 250 °C	Annealed at 300 °C
0 sccm	-	129.6	158.8
2.5 sccm	-	63.4	113.8
5.0 sccm	-	74.2	75.5

Table III. Calculated grain size for crystallized ITO films.

Grain size [nm]	Annealed at 200 °C	Annealed at 250 °C	Annealed at 300 °C
0 sccm	-	27.2	25.6
2.5 sccm	-	17.6	22.1
5.0 sccm	-	22.4	21.7

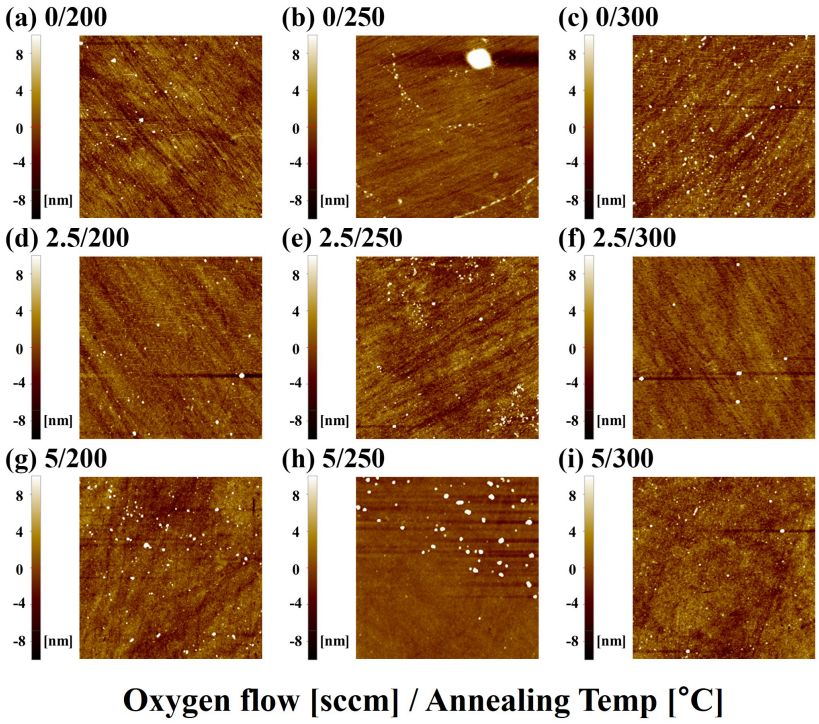


Figure 7. Two-dimensional AFM surface morphology images of ITO thin films deposited under various conditions. AFM images were obtained using a Park NX10 in non-contact mode over a 5 × 5 μm^2 scan area with a resolution of 256 × 256 pixels. (a–c) Images of films deposited at an O_2 flow rate of 0 sccm and annealed at 200, 250, and 300 °C, respectively. (d–f) Images of films deposited at an O_2 flow rate of 2.5 sccm and annealed at 200, 250, and 300 °C, respectively. (g–i) Images of films deposited at an O_2 flow rate of 5 sccm and annealed at 200, 250, and 300 °C, respectively. Color bar: height scale in nm.

fully explained by crystallinity alone and is attributed to a possible increase in surface roughness resulting from excessive crystallization along the (222) direction. This interpretation is further supported by AFM analysis, as the I_s is proportional to the square of the surface roughness, according to the Debye–Waller factor Eq. (1) [12,13].

The XRD analysis results also provide important insights into the changes in sheet resistance. As shown in Fig. 4 and Table III, under 0 sccm conditions, the sheet resistance decreased dramatically by approximately one order of magnitude when the annealing temperature increased from 200 to 250 °C. Previous studies have reported that when a high-quality crystalline phase is not formed in ITO thin films, the defect density at grain boundaries increases, which can severely hinder charge carrier transport and lead to an increase in film resistance [29]. In the present study, the 0 sccm sample annealed at 250 °C exhibited the highest crystallinity and the largest grain size, indicating that the charge transport pathways were significantly improved due to a reduction in grain boundary density and defect concentration. Therefore, the rapid decrease in sheet resistance under 0 sccm conditions can be interpreted as a consequence of the superior crystal quality and enhanced grain growth confirmed by XRD analysis.

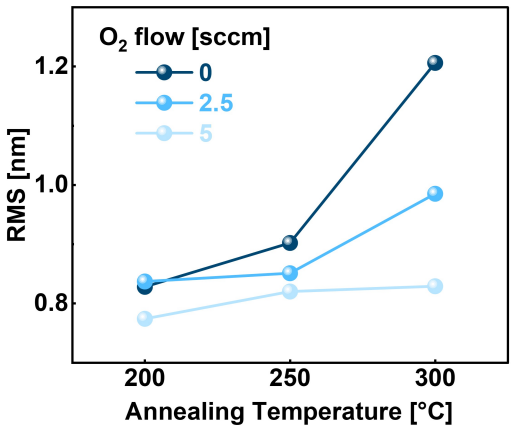


Figure 8. RMS surface roughness of ITO thin films deposited at O_2 flow rates of 0, 2.5, and 5 sccm as a function of annealing temperature (200, 250, and 300 °C).

4.6. Atomic force microscopy analysis

The surface morphology and roughness were analyzed using AFM. Figure 7 shows the surface morphology of all samples, and Fig. 8 summarizes the trend of surface roughness values as a function of annealing temperatures, for 0, 2.5, and 5.0 sccm of O_2. One can notice that while other samples showed a modest 5 – 10 % increase in roughness with a 50 °C temperature rise, the 0 sccm sample exhibited a sharp jump of over 30 % in RMS roughness when the temperature was increased from 250 to 300°C. This finding can explain a decrease in transmittance for O_2-0-sccm samples, when the annealing temperature increased from 250 to 300°C (Fig. 3).

Debye-Waller factor Eq. (1) states that the I_s is proportional to the square of the surface roughness. Therefore, the 30 % increase in roughness for the 0 sccm, 300 °C sample led to a significant increase in light scattering. In this specific case, the negative impact of increased roughness on transmittance became the dominant factor, overriding the positive effect of further crystallization. This explains why the transmittance decreased despite the higher annealing temperature. In essence, the benefits of further crystallization were outweighed by the detrimental effects of increased surface roughness, which led to dominant light scattering.

5. Conclusions

This study successfully optimized the fabrication conditions of ITO thin films by systematically investigating the effects of O_2 flow rate during sputtering and post-deposition annealing temperature. It was confirmed that a lower O_2 flow rate promotes the formation of O_2 vacancies, leading to enhancement in electrical conductivity, while increasing the annealing temperature promotes crystallization, which generally improves both electrical conductivity and optical transmittance.

For an overall evaluation of optoelectronic performance, Haacke's FOM was employed to quantitatively assess the balance between transmittance and conductivity. As shown in Fig. 5, the FOM analysis confirms that an O_2 flow rate of 0 sccm represents the optimal processing condition, and that the balance between transparency and conductivity is maximized at annealing temperatures of 250 °C or higher. In particular, the FOM values of the samples deposited at 0 sccm and annealed at 250 and 300 °C were more than one order of magnitude higher than those obtained under all other conditions, indicating superior overall optoelectronic performance.

From the electrical perspective, the sheet resistance decreased sharply with increasing annealing temperature; However, when the O_2 flow rate was 0 sccm, the rate of sheet resistance reduction was significantly suppressed at annealing temperatures above 250 °C, indicating a saturation behavior. From the optical perspective, the transmittance decreased at annealing temperatures of 250 °C or higher only when the O_2 flow rate was 0 sccm, due to excessive crystallization. These results reveal a clear trade-off between crystallization and surface roughness in ITO thin films annealed above 250 °C at an O_2 flow rate of 0 sccm. In particular, the rapid increase in surface roughness caused by excessive crystallization becomes the dominant factor governing optical transmittance by significantly enhancing light scattering.

Therefore, 250 °C can be identified as a critical annealing temperature at which excessive crystallization begins to adversely affect optical performance. Based on these findings, an O_2 flow rate of 0 sccm and an annealing temperature of 250 °C or higher are identified as the optimal processing conditions for ITO thin films. The results of this study provide valuable insights for optimizing process parameters in transparent thin-film transistor applications.

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Conflicts of Interest

The authors declare no conflicts of interest.

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